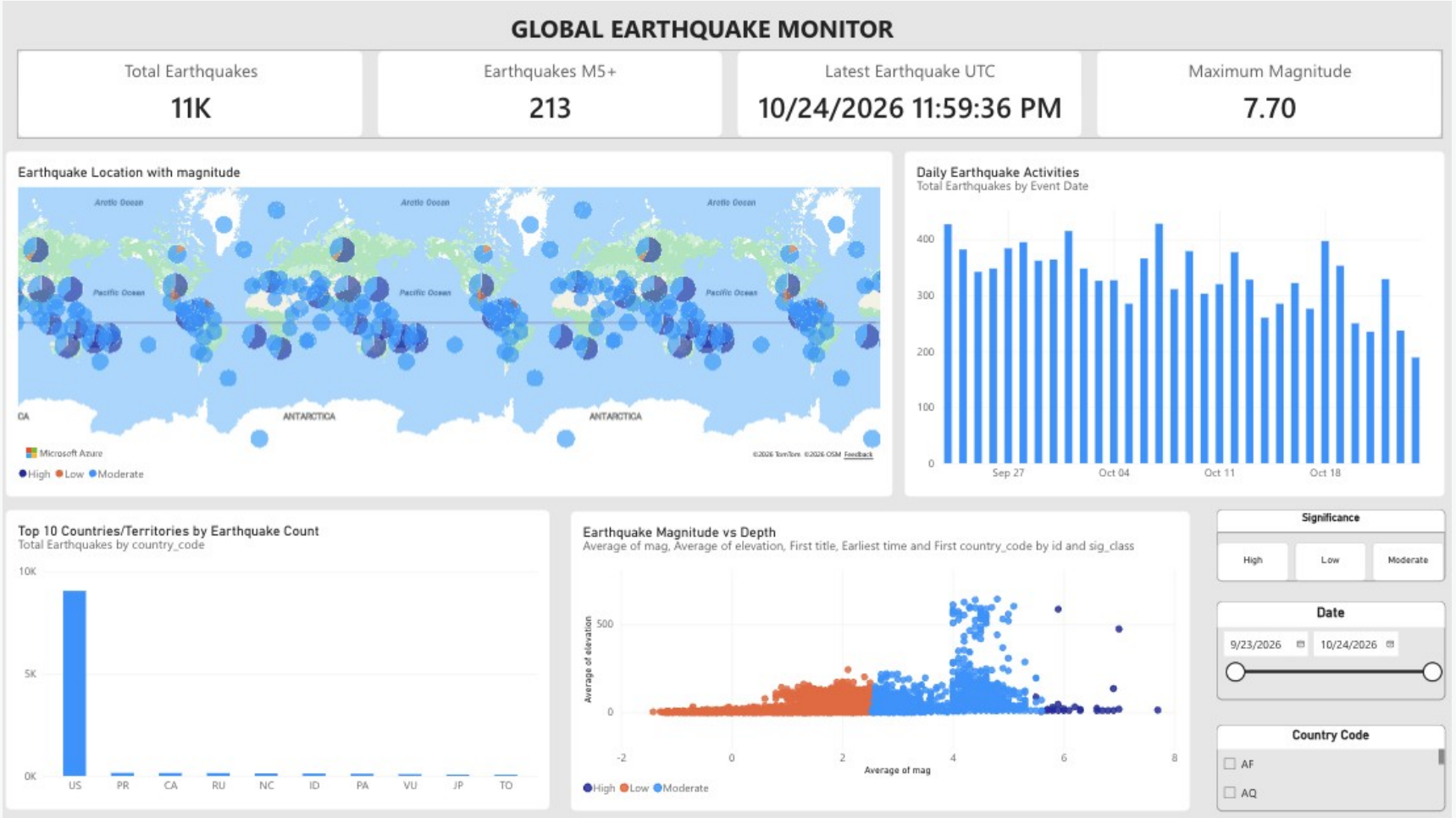
Global Earthquake Monitor

An automated lakehouse pipeline
with an interactive Power BI report

USGS API / Microsoft Fabric / Power BI

Global earthquake activity in one view



Report snapshot: 23 Sep–24 Oct 2026

Key figures for the selected period

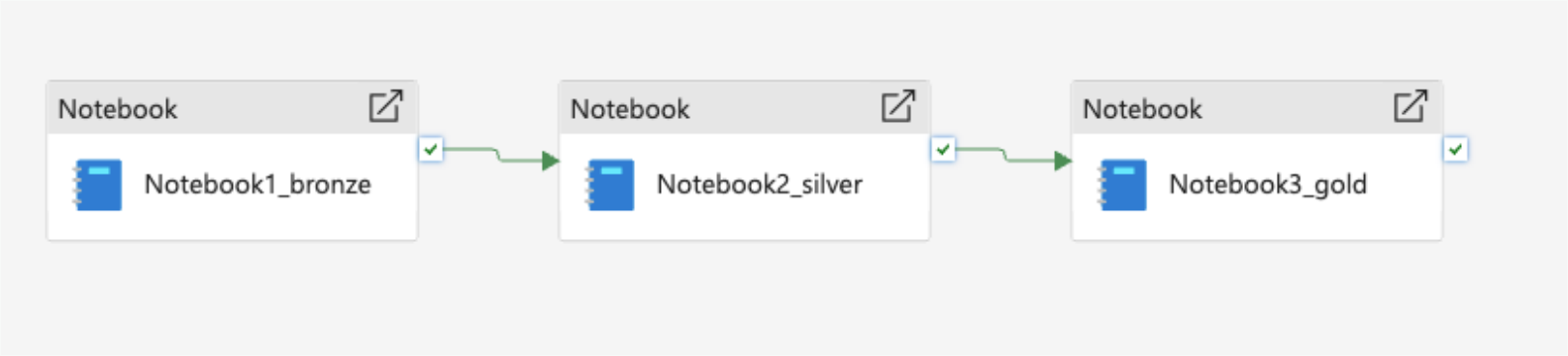
≈11K	213	7.70
Earthquakes	**Magnitude 5+ events**	**Maximum magnitude**

Latest event in the report

24 October 2026, 23:59:36 UTC

The total is rounded in the dashboard. All figures reflect the visible filters.

Bronze, Silver and Gold in Fabric



01 Bronze

Call the USGS API
Save JSON files by date range

02 Silver

Flatten GeoJSON fields
Convert event timestamps
Write a Delta table

03 Gold

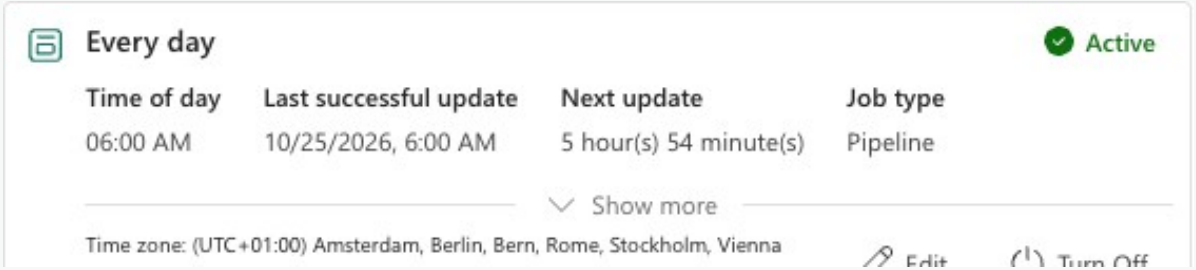
Add country codes
Classify significance
Merge by earthquake ID

Source: USGS GeoJSON API Reporting: Gold v2, Direct Lake and Power BI

30 days initially, then one day per run

Initial load

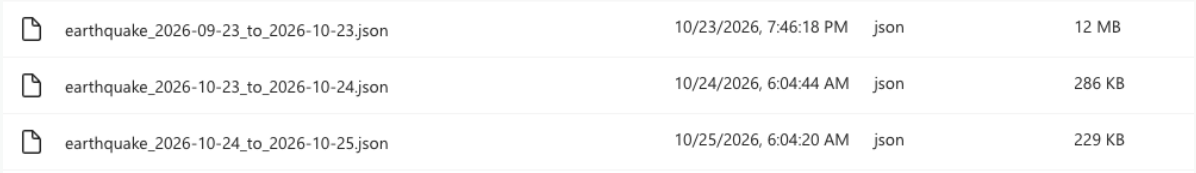
One file holds the historical
30-day date range.



Daily load

The pipeline retrieves the previous
complete UTC day at 06:00
in the Berlin schedule time zone.

Date ranges in the Bronze filenames



Daily ingestion adds to the historical dataset used by the report.

Duplicate handling in the Gold layer

The problem

Appending overlapping API results can create several rows for the same earthquake.

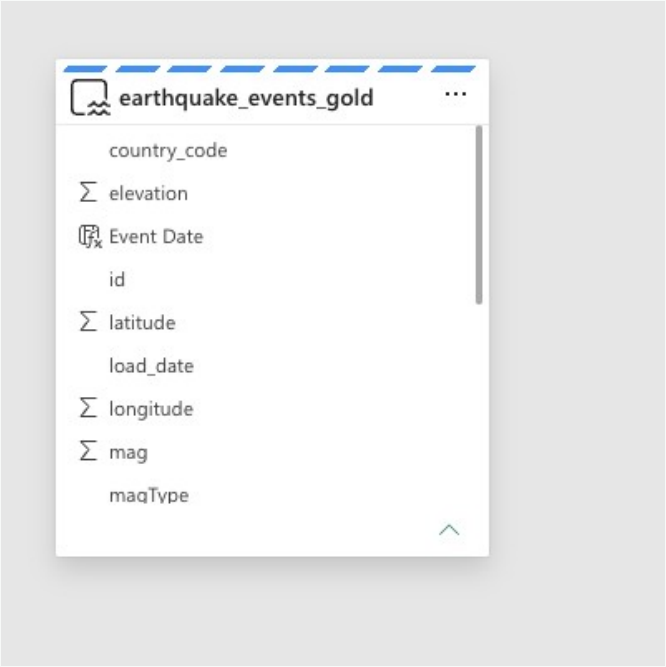
The change

Keep the newest incoming row per ID.
Merge into Delta using the event ID.
Update only when the source is newer or has the same updated timestamp.

MERGE key: target.id = source.id

Validation: row count vs distinct IDs, duplicate groups and event date boundaries

Direct Lake connects Gold to Power BI



KPI measures

Distinct event count, M5+ count,
maximum magnitude and latest event UTC

Analysis views

Map for location patterns
Daily counts for activity over time
Country ranking and magnitude vs depth

Slicers: Event Date, significance and country code

Tools behind the project

Fabric Data Factory

Schedule the three notebook activities

OneLake and Lakehouse

Store JSON files and Delta tables

Python and Requests

Retrieve data from the USGS API

PySpark and Delta Lake

Transform records and merge by ID

reverse_geocoder and Spark SQL

Add country codes and validate data

Power BI, Direct Lake and DAX

Model the data and build the report

Next improvement: capture late revisions to older events.